DRUGS · HEAD AND NECK ONCOLOGY

Pembrolizumab and nivolumab

Pembrolizumab and nivolumab are antibodies against programmed death receptor 1 (PD-1). By blocking the brake on cytotoxic T cells, they let the immune system attack the tumour. Pembrolizumab, alone or with chemotherapy, is first-line treatment for recurrent or metastatic head and neck squamous cell carcinoma expressing PD-L1 (KEYNOTE-048); nivolumab improved survival after platinum failure (CheckMate 141); and perioperative pembrolizumab has improved event-free survival in resectable locally advanced disease (KEYNOTE-689). Their side effects are immune attacks on normal organs, which can appear months after starting.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

PEMBROLIZUMAB AND NIVOLUMAB

	SUMMARY
Class	Immune checkpoint inhibitors (anti-PD-1)
How they work	Block PD-1 on T cells so tumour PD-L1 cannot switch them off
ENT uses	Recurrent or metastatic HNSCC first-line (pembrolizumab alone if PD-L1 combined positive score of 1 or more, or with platinum and 5-FU); after platinum failure (nivolumab, pembrolizumab); perioperative treatment of resectable locally advanced HNSCC (pembrolizumab, approved in some countries); cutaneous squamous cell carcinoma (cemiplimab); nasopharyngeal carcinoma (toripalimab and others with chemotherapy)
Dose	Pembrolizumab 200 mg every 3 weeks or 400 mg every 6 weeks; nivolumab 240 mg every 2 weeks or 480 mg every 4 weeks
Biomarker	PD-L1 **combined positive score** (CPS) on tumour tissue

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Response patterns

Responses can be slow and durable; pseudoprogression (apparent growth from immune infiltration) occurs occasionally, so scans are interpreted in context. About a fifth of patients respond, and some have long-term control.

3 CHAPTER THREE Safety

Immune-related adverse events.

3.1 Recognising them

Any organ can be affected:

- **Thyroid**: hypothyroidism (very common, particularly after neck radiotherapy) or thyroiditis with transient thyrotoxicosis; check thyroid function regularly.
- **Skin**: rash, pruritus, vitiligo.
- **Gut**: diarrhoea and colitis.
- **Lung**: pneumonitis (cough, breathlessness).
- **Liver**: hepatitis.
- **Endocrine**: hypophysitis, adrenal insufficiency, type 1 diabetes.
- Rarer: myocarditis (often fatal), myositis, neuropathy, nephritis, sicca.

3.2 Management

Grade the toxicity, withhold the drug for moderate or severe events, and treat with corticosteroids (with other immunosuppressants for refractory cases) in

discussion with oncology; replace hormones for endocrine failure, which is usually permanent.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (PD-1 and PD-L1) and the **CPS** biomarker.
- **KEYNOTE-048** and CheckMate 141.
- Common immune-related adverse events, especially thyroid.

4.2 Common pitfalls

COMMON PITFALLS

- Missing hypothyroidism or adrenal insufficiency in a tired patient on immunotherapy.

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